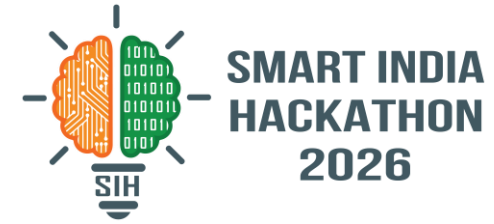
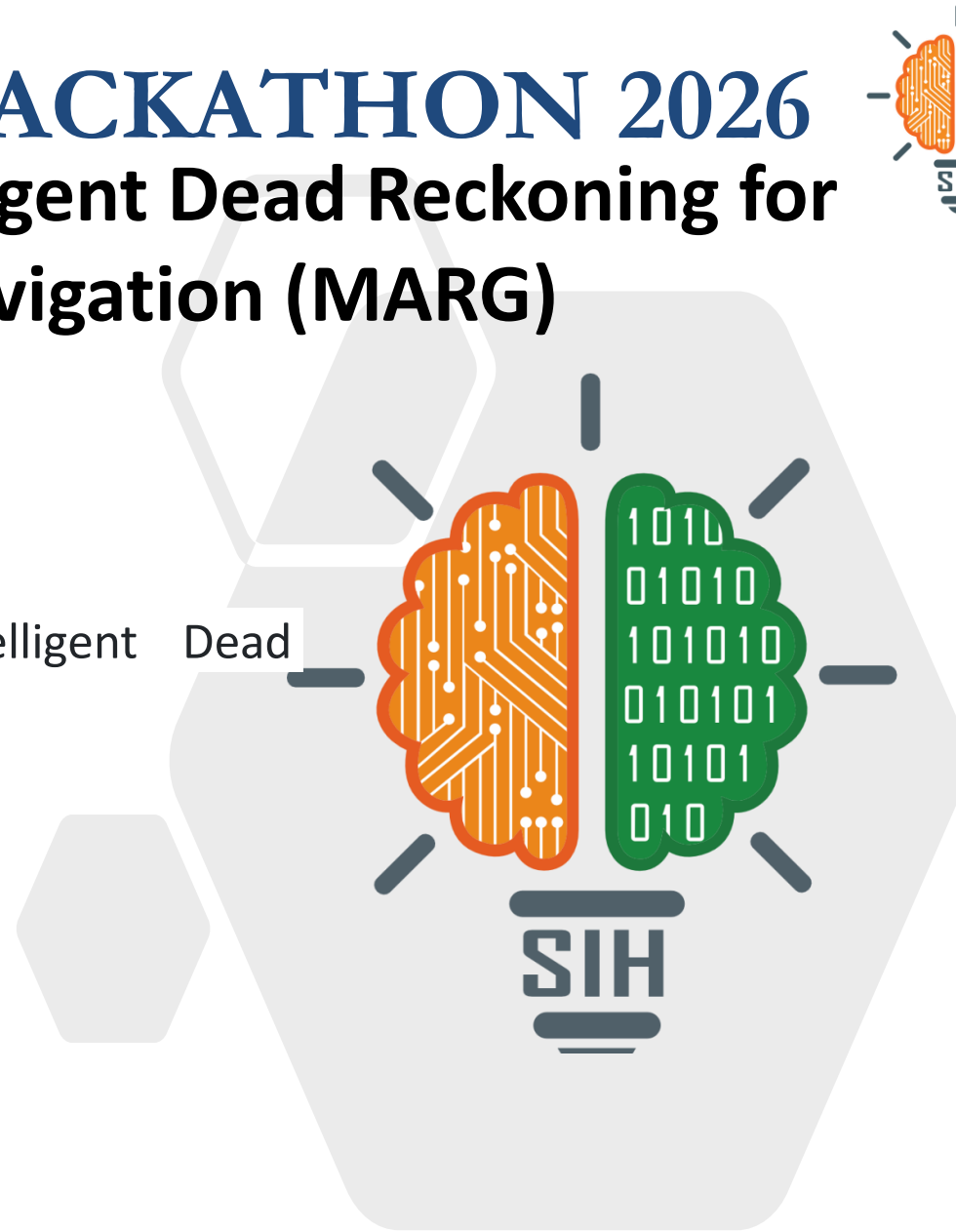


SMART INDIA HACKATHON 2026

Uncertainty-Gated Intelligent Dead Reckoning for GNSS-Denied Navigation (MARG)



- **Problem Statement ID-** SIH26168
- **Problem Statement Title-** AI-ML based Intelligent Dead Reckoning system for seamless navigation
- **Theme-** Smart Vehicles
- **PS Category-** Software
- **Team ID-** 150293
- **Team Name -** BlitzKrieg



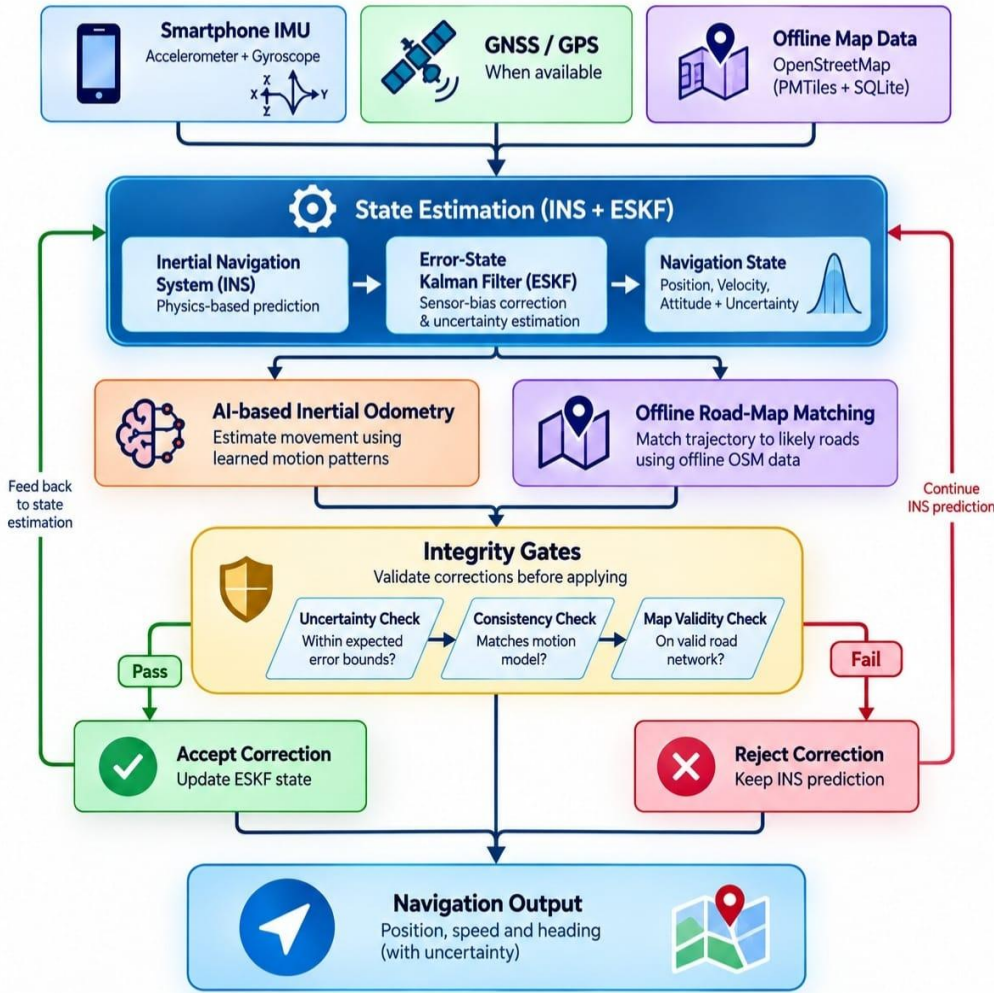
[GitHub Repo\(Click Me\)](#)

How it addresses the problem:

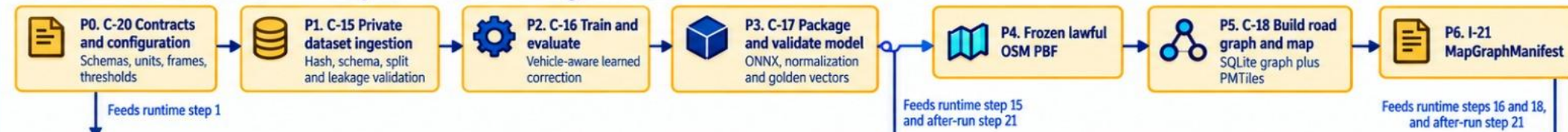
- Ensures **continuous navigation during GNSS/GPS loss in tunnels, urban canyons, dense foliage, multipath zones, and jamming environments** by automatically switching to **Dead Reckoning (DR)**.
- Combines **IMU + INS/ESKF + AI Inertial Odometry + Offline Road Matching** to estimate **position, velocity, and heading**, while reducing **drift** and tracking **navigation uncertainty**.
- Uses strict **Integrity Gates** and **controlled GNSS Reacquisition** with **Age, Accuracy, Innovation/NIS & Consistency Checks**, preventing unreliable corrections and sudden **position jumps** when GNSS returns

Innovation and uniqueness of the solution:

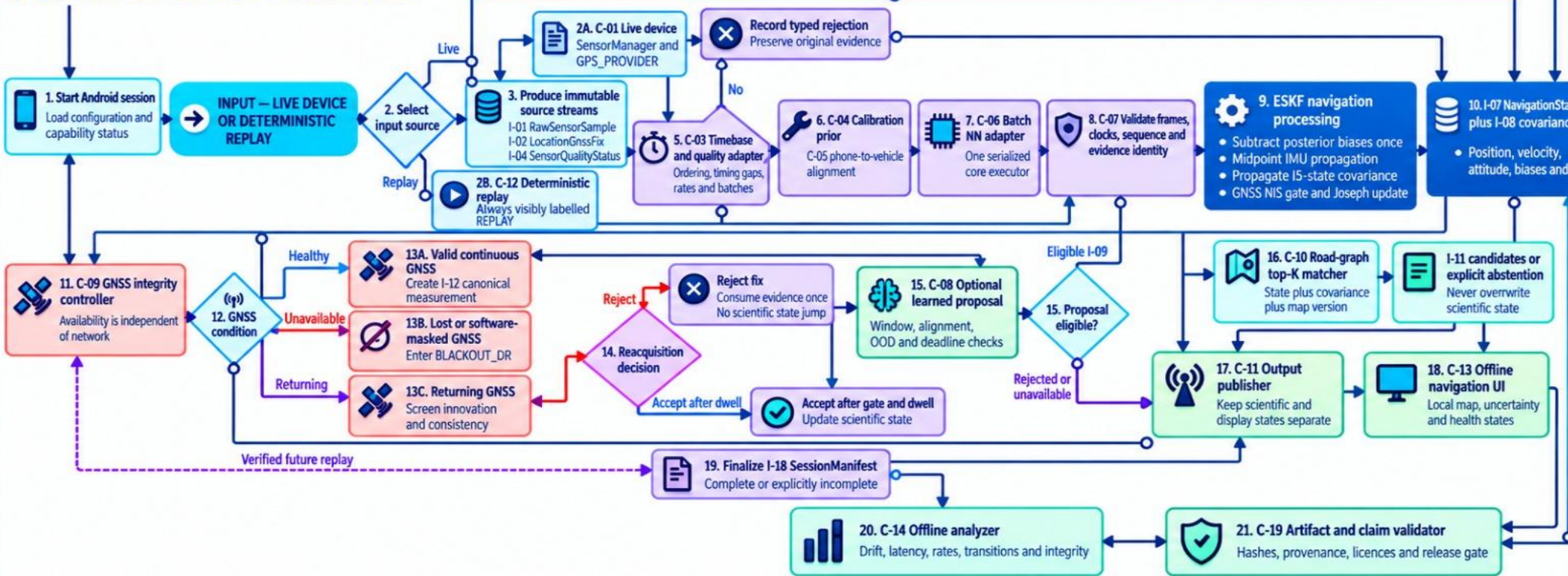
- Hybrid Architecture:** Integrates **Classical INS + ESKF + AI Inertial Odometry + Offline Road Matching + Integrity Monitoring** for accurate and resilient navigation.
- Safe AI Integration:** Keeps the **Classical Navigation Engine** as the **Single State Authority**; AI/map corrections are accepted only after **OOD, Confidence & Integrity Gates**.
- Intelligent Map Matching:** Uses **Top-K Road Hypotheses with Abstention**, avoiding forced or incorrect road matches during uncertainty.
- Graceful Degradation:** AI failure → **Classical DR**, Map failure → **Unmatched DR**, Bad GNSS → **Continue DR**, Critical fault → **FAULT**, making the system **Offline-First, Explainable, Fail-Safe & Robust**.



OFFLINE PREREQUISITES — completed before a navigation run



SCIENTIFIC RUNTIME — execution order

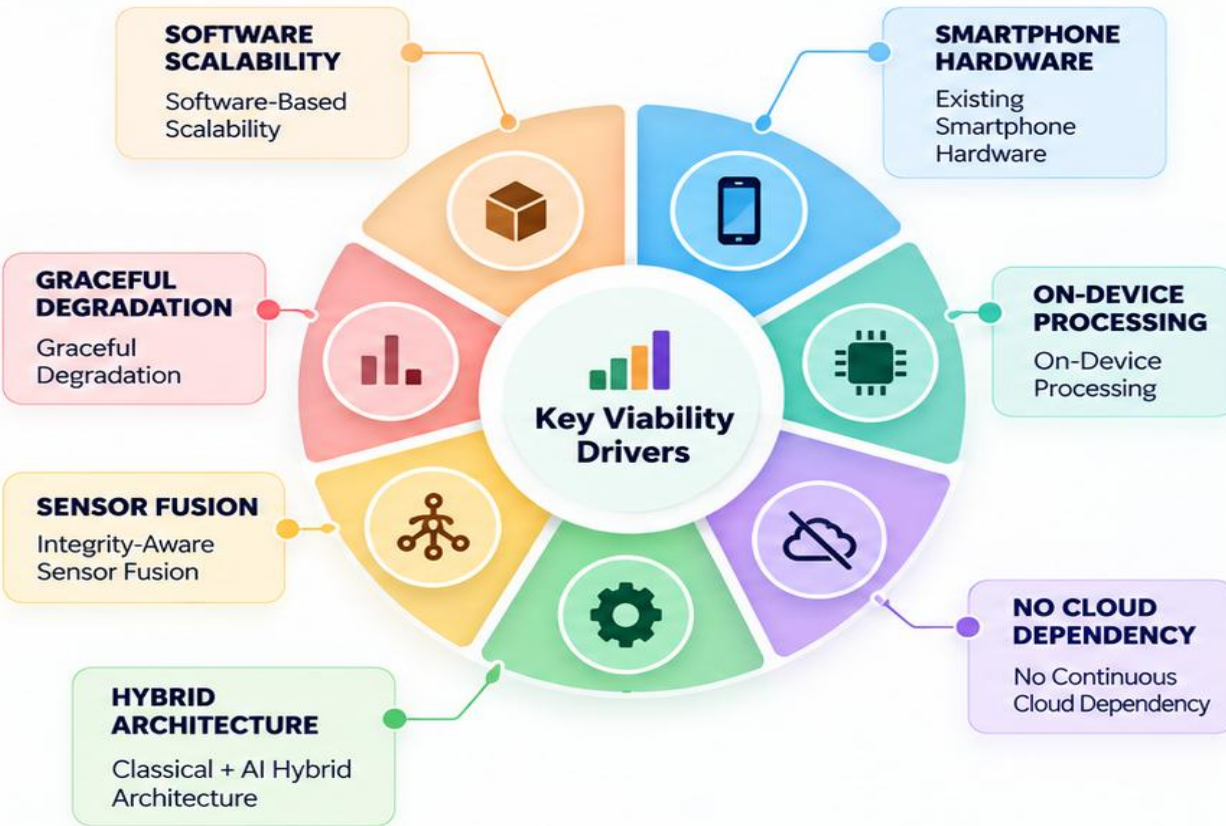


Technology Stack

On-Device | Offline | Open Source

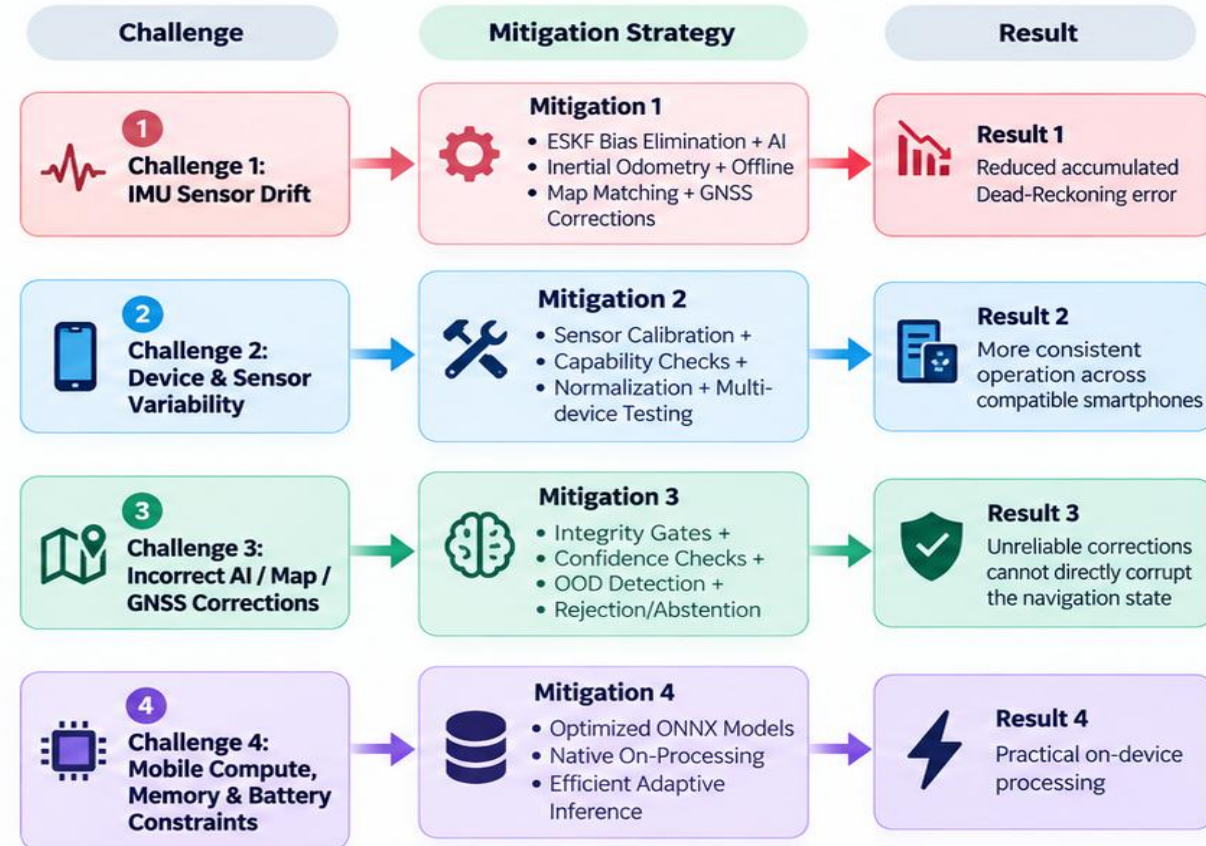


SOLUTION VIABILITY ASSESSMENT



CHALLENGES AND MITIGATION STRATEGIES

FLOWCHART





GNSS-CHALLENGED ENVIRONMENTS

Tunnels, urban canyons, dense foliage, and multipath interference cause satellite signals to be blocked, obstructed, or degraded, leading to GNSS degraded or lost.

DEAD RECKONING ACTIVATED

Smartphone IMU, AI Inertial Odometry, and Offline Map Matching feed into the INS and ESKF navigation core to produce position, velocity, heading, and uncertainty.

CONTINUOUS NAVIGATION DURING GNSS OUTAGES

Ensures no navigation freeze, reduced dependence on GNSS, integrity-aware positioning, and reliable offline operation.

TARGET AUDIENCES IMPACT

Provides navigation continuity for drivers, emergency services, mobility applications, and a research platform for developers.

*"Navigation that doesn't stop,
so people don't either."*

1. Smartphone GNSS & Sensor-Based Navigation

- Google / Android Developers (2026). [Raw GNSS Measurements on Android Devices](#) — Supports the feasibility of accessing **raw** GNSS measurements on Android smartphones.
- Android Developers (2026). [GNSS Raw Measurement Analysis using GnssLogger](#) — Reference for **GNSS measurement validity, uncertainty and signal-quality analysis**.

2. AI-Based Inertial Dead Reckoning

- AVNet Researchers (2025). [AVNet: Learning Attitude and Velocity for Vehicular Dead Reckoning Using Smartphone](#) — Demonstrates **deep learning + Kalman filtering + smartphone IMU** for vehicle dead reckoning in **GNSS-denied environments**.

3. INS & Map-Matching in GNSS-Denied Environments

- Wang, X., Gilliam, C., Kealy, A., et al. (2023). [Probabilistic Map Matching for Robust Inertial Navigation Aiding](#) — Demonstrates **multiple-hypothesis map matching** for reducing accumulated **INS drift and bias**.
- Land Vehicle Localization Study (2026). [Land Vehicle Localization in GNSS-Denied Zones Using IMU and Map with Terrain Features](#) — Supports combining **IMU + road-map information** for vehicle localization when GNSS is unavailable.

4. GNSS Reliability & Integrity

- Android Developers (2026). [GnssMeasurement API Reference](#) — Documents **measurement state, SNR/CN0, accumulated delta range and measurement uncertainty** useful for GNSS integrity checks.
- Android Developers – Research Publications. [GNSS Positioning Research & Publications](#) — Collection of research covering **smartphone GNSS positioning, multipath mitigation, raw measurements and positioning limitations**.